Mobile Computing Overview

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H. Stormer, ZHAW, T. Amberg, FHNW

Slides: tmb.gr/mc-how



Source on GDocs

Overview

These slides make sure you know what's up:

What you can expect from this course.

What is expected from you.

Hello

Prof. Thomas Amberg, thomas.amberg@fhnw.ch, lecturer for IoT at FHNW, Institute of Mobile and Distributed Systems.

Prof. Dr. B. Fehrensen, benjamin.fehrensen@bfh.ch, lecturer for IT-Security and Privacy at BFH, Institute for Cybersecurity and Engineering.

ECTS and effort

3 ECTS credits => 90 hours:

14 sessions à 3 hours = 42 hours in class.

+ 24 extra hours for an integrated team project.

+ 24 hours of private studies & exam preparation.

Learning objectives: app development

Students know how to design and implement native applications for mobile devices running Android.

Students have basic knowledge of UI design guidelines and techniques relevant for mobile app design.

Students can describe the integration of their application with a backend Web service.

Learning obj.: sensors & connectivity

Students know how to use smartphone on-device sensor APIs for motion, position and environment.

Students have basic knowledge of prototyping a peripheral IoT device with an Arduino-compatible hardware platform, sensors and actuators.

Students have basic knowledge of connectivity options like Bluetooth Low Energy (BLE).

Dates 2026

14.09.	Introduction & App. Development 1
21.09.	Android Application Development 2
28.09.	Android Application Development 3
05.10.	Android Application Development 4
12.10.	Sensors & Connectivity 1
19.10.	Sensors & Connectivity 2
26.10.	Sensors & Connectivity 3
02.11.	Sensors & Connectivity 4
…	

Dates 2026/27

09.11.	Project Kick-off
16.11.	Independent Project Work 1
23.11.	Independent Project Work 2
30.11.	Independent Project Work 3
07.12.	Independent Project Work 4
14.12.	Project Presentation
W52/53	(no class)
W1-4	(no class)
W5-6	Assessment (TBA)
W10/11	Results (on demand)

Learning objective assessment

A graded team project, due on *Presentation Day 0:00*.

A mandatory, written assessment of 120 minutes.

Counting 30% (project) and 70% (assessment).

The final grade will be a Bologna grade.

Assessment

120 minutes, open-book, written assessment.

Slides and a personal summary are allowed.

Course repositories and Web search, too.

No communication (phone, chat*, …).

*Including AI tools.

Team project

2 or 3-student project teams, building an IoT system.

8' online presentation of the project, plus a 2' video.

Slides, source code, binaries, README on GitHub.

All team members are able to explain the project.

Details at the project kick-off.

Plagiarism

Unfortunately has to be mentioned, sanctions apply.

From *Betrug und Plagiate bei Leistungsnachweisen*:

"Wer in Arbeiten im Rahmen des Studiums Eigen- und Fremdleistung nicht unterscheidet, wer plagiiert, macht sich strafbar." - M. Meyer

Using 3rd-party code? Make it clear, check license.

Course materials

The official course site is *TSM_MobCom* on Moodle.

Android content will be there, sensors & connectivity

slides and source code examples are on GitHub at

https://github.com/tamberg/mse-tsm-mobcom

Communication

We use Microsoft Teams to communicate.

For remote access via video stream.

And as a asynchronous chat.

Tools and languages

Android Studio for Android native development.

Arduino IDE for Arduino C development.

Git version control for source code.

AI tools are not required.

Smartphone

Bring your own Android smartphone.

Ideally not much older than 3 years.

Android version 9, or newer.

No such phone? Get in touch.

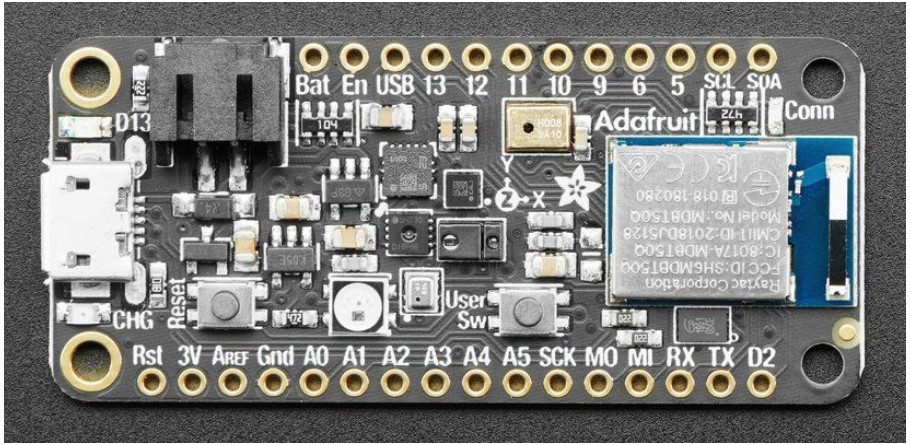
Embedded hardware

Adafruit Feather nRF52840 Sense

ARM Cortex M4 based SoC

Bluetooth Low Energy

Built-in sensors



We'll lend you one.

CC BY-SA, adafruit.com

Feedback or questions?

Join us on *MSE TSM MobCom* in MS Teams*

Or email thomas.amberg@fhnw.ch and

benjamin.fehrensen@bfh.ch

Thanks for your time.